VARUN SAXENA

varunsaxena314@gmail.com • 703-851-2796 • linkedin.com/in/varun-saxena314 • https://github.com/Varun-Saxena314

EDUCATION

University of Virginia | School of Arts and Sciences — GPA: 3.813
M.S in Computer Science — May 2027

SKILLS

Coding Proficiencies and Frameworks: Java, Python, Javascript, HTML, CSS, C, SQL, React.JS, PyTorch, TensorFlow, Flask
Knowledge/Technologies: Git, Github, Terminal, Visual Studio Code, Eclipse, MySQL, Firebase, Azure OpenAI, Microsoft 365

EXPERIENCE

ICF International | *AI Software Developer Intern* — Jun. 2025-Aug. 2025

- Designed internal **AI-driven automation tools** to streamline creation, review, and evaluation of government contracts and RFP responses for consultants and program managers
- Integrated **OpenAI agents** into Microsoft 365 using **Azure AI Foundry, Copilot, and Python**, automating document parsing, summarization, and proposal scoring
- Boosted operational efficiency of contract workflows by **40%**, reducing manual labor and accelerating proposal turnaround times across multiple business units

UVA School of Engineering | *Computer Systems and Organizations 1 TA* — Jan. 2025-Present

- Assisted with design and refinement of course content, including lectures, homework, and exams
- Held office hours and assisted in in-person lectures to 100+ students, teaching the fundamentals of binary, circuits, machine code, assembly, and **C programming**
- Achieved class averages of over **80%** on exams and over **85%** on homework, demonstrating improved student understanding

Engauge | *Machine Learning and AI Intern* — Aug. 2024-Jan. 2025

- Developed a **spoken language processing AI** that searched through student comments and questions during college lectures to categorize topics, enabling professors to efficiently address critical questions in real time
- Utilized **Python's NLTK** library to parse pre-categorized comments from 20+ college lectures, training a model to accurately organize future lecture discussions based on relevance
- Deployed the AI model in **15+** STEM courses with **2,000+** students at the University of Maryland, resulting in a **30%** increase in student engagement and **25%** improvement in instructor responsiveness

UVA Machine Learning and Artificial Intelligence | *Research Assistant* — Aug. 2024-Jun. 2025

- Created a new generative AI chatbot which incorporates strengths from existing models to provide a more accurate and seamless discussion within the user-program interaction while lowering processing and maintenance costs
- Refined an open source **Large Language Model (LLM)** which is more accessible to operate while maintaining accuracy and specialized nature of responses found in closed source LLMs
- Achieved a **70% reduction** in operational cost while maintaining benchmark model performance, validated through automated regression tests

PROJECTS

Infracool | *Programming Lead* — Mar. 2025-Apr. 2025

- Directed development of an **Urban Heat Island Risk Predictor**, offering actionable and cost-efficient heat mitigation strategies based on city or ZIP code for the LMI-UVA Data Entrepreneurship Challenge
- Implemented a responsive **Flask-based** web app with a **Python backend** for real-time data analysis and user interaction
- Awarded **1st place** out of 10+ teams for empowerment and technical excellence with data in a university-wide competition

Portfolio Website | *Designer and Creator* — Jun. 2024

- Engineered a personal portfolio website, accomplishing seamless and dynamic design elements
- Employed the **React.JS framework**, along with **Javascript, CSS, and HTML** to display an engaging frontend design

Nth Degree Robotics Team | *Autonomous Lead* — Aug. 2022-Jun. 2023

- Conceptualized and created an **image processing system**, distinguishing different picture designs and recognizing color patterns, allowing for dynamic autonomous movement based off random pattern signals
- Applied **OpenCV/TensorFlow** Image Processing, allowing for a **50%** accuracy increase in color recognition
- Allowed **50% increase** in overall scoring in FTC robotic competitions, placing **top 4 in Virginia** for autonomous scoring

RELEVANT COURSEWORK

Classes: Data Structures and Algorithms 1 & 2, Computer Systems and Organizations 1 & 2, Discrete Mathematics 1 & 2, Introduction to Machine Learning, Software Development Essentials, Intro to Cybersecurity